EXERCISE-I (Conceptual Questions) Build Up Your Understanding

INTRODUCTION

1. Thermodynamics is concerned with :-
(1) Total energy of a system
(2) Energy changes in a system
(3) Rate of a chemical change
(4) Mass changes in nuclear reactions

TH0001

2. A well stoppered thermos flask contains some ice cubes. This is an example of :–
(1) Closed system
(2) Open system
(3) Isolated system
(4) Non-thermodynamic system

TH0002

3. Identify the intensive quantities from the following -
(1) Enthalpy and temperature
(2) Volume and temperature
(3) Enthalpy and volume
(4) Temperature and refractive index

TH0003

4. Which of the following is an extensive property
(1) Mass (2) Enthalpy
(3) Energy (4) All of these

TH0004

5. For an adiabatic process which of the following relations is correct
(1) $\Delta E = 0$ (2) $P\Delta V = 0$
(3) $q = 0$ (4) $q = + W$

TH0005

6. In which of the following process work is independent of path :
(1) Isothermal (2) Isochoric
(3) Adiabatic (4) Isobaric

TH0006

7. When a gas is compressed adiabatically and reversibly, the final temperature is-
(1) Higher than the initial temperature
(2) Lower than the initial temperature
(3) The same as initial temperature
(4) Dependent upon the rate of compression

TH0007

8. Which one is dependent on initial and final state?
(1) Heat supplied at constant pressure
(2) Heat supplied at constant volume
(3) Enthalpy
(4) All of the above

TH0008

9. Out of boiling point (I), entropy (II), pH (III) and emf of a cell (IV), intensive properties are :
(1) I, III, IV (2) I, II
(3) I, II, III (4) All of these

TH0009

10. The work done by a weightless piston in causing an expansion ΔV (at constant temperature), when the opposing pressure P is variable, is given by :
(1) $W = -\int PdV$ (2) $W = 0$
(3) $W = -P\Delta V$ (4) None

TH0010

11. The work done by 100 calorie of heat in isothermal expansion of ideal gas is :-
(1) 418.4 J (2) 4.184 J
(3) 41.84 J (4) None

TH0011

12. Temperature and volume are not :-
(1) Extensive properties
(2) Intensive properties
(3) Intensive and extensive properties respectively
(4) Extensive and intensive properties respectively

TH0012

13. $q = -w$ is not true for :-
(1) Isothermal process (2) Adiabatic process
(3) Cyclic process (4) 1 and 3 both

TH0013

14. The temperature of an ideal gas increase in an -
(1) Adiabatic compression (2) Adiabatic expansion
(3) Isothermal expansion (4) Isothermal compression

TH0014

15. Which statement is true for reversible process :–
(1) It takes place in single step
(2) Driving force is much greater than opposing force
(3) Work obtained is minimum
(4) None

TH0016

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FIRST LAW OF THERMODYNAMICS ($\Delta E = q + W$)

16. Both q & w are_ _ _ _ _ function :-
(1) State (2) State, Path
(3) Path, State (4) Path
TH0017

17. If work done by the system is 300 joule when 100 cal. heat is supplied to it. The change in internal energy during the process is :-
(1) – 200 Joule (2) 400 Joule
(3) 720 Joule (4) 120 Joule
TH0018

18 A system has internal energy equal to E_1, 450 J of heat is taken out of it and 600 J of work is done on it. The final energy of the system will be -
(1) (E_1 + 150) (2) (E_1 + 1050)
(3) (E_1 – 150) (4) None of these
TH0019

19. The work done by a system is 8J when 40J heat is supplied to it. The change in internal energy of the system during the process :
(1) 32 J (2) 40 J
(3) 48 J (4) –32 J
TH0020

ENTHALPY [$\Delta H = \Delta E + P\Delta V/\Delta H = \Delta E + \Delta n_g RT$]

20. Internal energy change during a reversible isothermal expansion of an ideal gas is :-
(1) Always negative
(2) Always positive
(3) Zero
(4) May be positive or negative
TH0022

21. Under which of the following conditions is the relation, $\Delta H = \Delta E + P\Delta V$ valid for a system :–
(1) Constant pressure
(2) Constant temperature
(3) Constant temperature and pressure
(4) Constant temperature, pressure and composition
TH0023

22. The difference between heats of reaction at constant pressure and constant volume for the reaction $2C_6H_6(l) + 15O_2(g) \longrightarrow 12CO_2(g) + 6H_2O(l)$ at 25°C in KJ is
(1) + 7.43 (2) +3.72
(3) – 7.43 (4) – 3.72
TH0024

23. For a gaseous reaction,
$A(g) + 3B(g) \longrightarrow 3C(g) + 3D(g)$
ΔE is 17 kCal at 27°C assuming R = 2 Cal K^{-1} mol^{-1}, the value of ΔH for the above reaction is:
(1) 15.8 Kcal (2) 18.2 Kcal
(3) 20.0 Kcal (4) 16.4 Kcal
TH0025

24. Which of the following statements is correct for the reaction ; $2SO_2(g) + O_2(g) \longrightarrow 2SO_3(g)$ at constant temperature and pressure
(1) $\Delta H = \Delta E$ (2) $\Delta H < \Delta E$
(3) $\Delta H > \Delta E$ (4) None of the above
TH0026

25. For the reaction $Ag_2O(s) \longrightarrow 2Ag(s) + \frac{1}{2}O_2(g)$, which one of the following is true :
(1) $\Delta H = \Delta E$ (2) $\Delta H = \frac{1}{2}\Delta E$
(3) $\Delta H < \Delta E$ (4) $\Delta H > \Delta E$
TH0027

26. A mixture of 2 moles of carbon monoxide and one mole of oxygen in a closed vessel is ignited to get carbon dioxide. If ΔH is the enthalpy change and ΔE is the change in internal energy, then :–
(1) $\Delta H > \Delta E$ (2) $\Delta H < \Delta E$
(3) $\Delta H = \Delta E$ (4) Not definite
TH0028

27. For the gaseous reaction involving the complete combustion of isobutane -
(1) $\Delta H = \Delta E$ (2) $\Delta H > \Delta E$
(3) $\Delta H = \Delta E = 0$ (4) $\Delta H < \Delta E$
TH0029

28. For the reversible isothermal expansion of one mole of an ideal gas at 300 K, from a volume of 10 dm^3 to 20 dm^3, ΔH is -
(1) 1.73 kJ (2) –1.73 kJ (3) 3.46 kJ (4) Zero
TH0030

29. For $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$ at 977°C, ΔH = 174 KJ/mol ; then ΔE is :-
(1) 160 kJ (2) 163.6 kJ
(3) 186.4 kJ (4) 180 kJ
TH0031

30. Heat of reaction for , $CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g)$ at constant V is –67.71 K cal at 17°C. The heat of reaction at constant P at 17°C is :-
(1) –68.0 kCal (2) + 68.0 kCal
(3) – 67.42 kCal (4) None
TH0032

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31. The enthalpy of vaporisation of water at 100°C is 40.63 kJ mol^{-1}. The value ΔE for this process would be:-
(1) 37.53 kJ mol^{-1} (2) 39.08 kJ mol^{-1}
(3) 42.19 kJ mol^{-1} (4) 43.73 kJ mol^{-1}

TH0034

32. For the system $S(s) + O_2(g) \rightarrow SO_2(g)$:–
(1) ΔH = ΔE (2) ΔH > ΔE
(3) ΔE > ΔH (4) ΔH = 0

TH0035

33. For the reaction $CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g)$

Which one of these statement is correct at constant T and P ?
(1) ΔH = ΔE
(2) ΔH < ΔE
(3) ΔH > ΔE
(4) ΔH is Independent of physical state of reactants

TH0036

34. Which is true for the combustion of sucrose ($C_{12}H_{22}O_{11}$) at 25°C :-
(1) ΔH > ΔE (2) ΔH < ΔE
(3) ΔH = ΔE (4) None

TH0037

35. For which change ΔH ≠ ΔE :-
(1) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$
(2) $HCl(\ell) + NaOH(\ell) \rightarrow NaCl(s) + H_2O(\ell)$
(3) $C(s) + O_2(g) \rightarrow CO_2(g)$
(4) $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

TH0038

36. The heat of combustion of ethanol determined in a bomb calorimeter is – 670.48 kCal $mole^{-1}$ at 27°C. What is ΔH at 27°C for the reaction :–
(1) – 335.24 kCal (2) – 671.08 kCal
(3) – 670.48 kCal (4) + 670.48 kCal

TH0039

37. The difference in ΔH and ΔE for the combustion of methane at 25°C would be :–
(1) Zero (2) 2 × 298 × – 2 Cals.
(3) 2 × 298 × – 3 Cals. (4) 2 × 25 × – 3 Cals.

TH0040

38. For which of the following reactions ΔH is less than ΔE :–
(1) $C_{12}H_{22}O_{11}(s) + 6O_2(g) \rightarrow 6CO_2(g) + 6H_2O(\ell)$
(2) $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$
(3) $N_2O_4(g) \rightarrow 2NO_2(g)$
(4) $N_2(g) + O_2(g) \rightarrow 2NO(g)$

TH0041

39. For a reaction $2X(s) + 2Y(s) \rightarrow 2C(\ell) + D(g)$
The q_p at 27°C is – 28 kCal mol^{-1}.
The q_V is ---------- kCal mol^{-1} :–
(1) – 27.4 (2) + 27.4 (3) – 28.6 (4) 28.6

TH0042

WORK DONE IN DIFFERENT PROCESS

40. The work (in ergs) for a reversible expansion of one mole of an ideal gas from a volume of 10 litres to 20 litres at 25°C is :
(1) $-2.303 \times 8.314 \times 10^7 \times 298 \log 2$
(2) $-2.303 \times 0.0821 \times 298 \log 2$
(3) $-2.303 \times 0.0821 \times 298 \log 0.5$
(4) $-2.303 \times 2 \times 298 \log 2$

TH0043

41. Two moles of an ideal gas expand spontaneously into vacuum. The work is :–
(1) Zero (2) 2 J
(3) 4 J (4) 8 J

TH0045

42. One mole of a gas occupying $3dm^3$ expands against a constant external pressure of 1 atm to a volume of 13 lit. Find work is :–
(1) – 10 atm dm^3 (2) – 20 atm dm^3
(3) – 39 atm dm^3 (4) – 48 atm dm^3

TH0046

ENTROPY/SECOND LAW OF THERMODYNAMICS

43. For which reaction from the following, Δ S will be maximum ?
(1) $Ca(s) + \frac{1}{2}O_2(g) \longrightarrow CaO(s)$
(2) $CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$
(3) $C(s) + O_2(g) \longrightarrow CO_2(g)$
(4) $N_2(g) + O_2(g) \longrightarrow 2NO(g)$

TD0047

44. An adiabatic reversible process is one in which :-
(1) Temperature of the system does not change
(2) The system is not closed to heat transfer
(3) There is no entropy change
(4) None of these

TD0048

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45. Entropy means
(1) Disorderness (2) Randomness
(3) Orderness (4) both 1 & 2
TD0049

46. ΔS for the reaction;
$MgCO_3(s) \longrightarrow MgO(s) + CO_2(g)$ will be :
(1) 0 (2) –ve (3) +ve (4) ∞
TD0050

47. Change in entropy is negative for
(1) $Br_2(\ell) \longrightarrow Br_2(g)$
(2) $C(s) + H_2O(g) \longrightarrow CO(g) + H_2(g)$
(3) N_2(g, 10 atm) $\longrightarrow$ N_2 (g, 1 atm)
(4) Fe(at 400 K) $\longrightarrow$ Fe(at 300 K)
TD0051

48. In which reaction ΔS is positive :-
(1) $H_2O(\ell) \rightarrow H_2O(s)$
(2) $3O_2(g) \rightarrow 2O_3(g)$
(3) $H_2O(\ell) \rightarrow H_2O(g)$
(4) $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$
TD0052

49. When the egg is hard boiled, there is-
(1) Increase in disorder
(2) Decrease in disorder
(3) No change in disorder
(4) ΔG is negative
TD0053

50. If S^0 for H_2, Cl_2 and HCl are 0.13, 0.22 and 0.19 kJ K^{-1} mol^{-1} respectively. The total change in standard entropy for the reaction
$H_2 + Cl_2 \longrightarrow 2HCl$ is :
(1) 30 JK^{-1} mol^{-1} (2) 40 JK^{-1} mol^{-1}
(3) 60 JK^{-1} mol^{-1} (4) 20 JK^{-1} mol^{-1}
TD0054

51. Which has the least entropy :
(1) Graphite (2) Diamond
(3) N_2(g) (4) N_2O(g)
TD0055

52. When two gases are mixed the entropy :–
(1) Remains constant (2) Decreases
(3) Increases (4) Becomes zero
TD0056

53. The enthalpy of vaporisation of per mole of ethanol (B.pt. = 79.5^0C and $\Delta S = 109.8$ JK^{-1} mol^{-1}) is :–
(1) 27.35 kJ mol^{-1} (2) 32.19 kJ mol^{-1}
(3) 38.70 kJ mol^{-1} (4) 42.37 kJ mol^{-1}
TD0058

54. If 900J/g of heat is exchanged at boiling point of water, then what is increase in entropy?
(1) 43.4 $JK^{-1}mole^{-1}$
(2) 87.2 $JK^{-1}mole^{-1}$
(3) 900 $JK^{-1}mole^{-1}$
(4) Zero
TD0059

55. 5 mole of an ideal gas expand reversibly from a volume of 8 dm^3 to 80dm^3 at a temperature of 27^0C. The change in entropy is :–
(1) 41.57 JK^{-1} (2) – 95.73 JK^{-1}
(3) 95.73 JK^{-1} (4) – 41.57 JK^{-1}
TD0060

56. In a spontaneous irreversible process the total entropy of the system and surroundings
(1) Remains constant (2) Increases
(3) Decreases (4) Zero
TD0061

57. The total entropy change for a system & its surroundings increases if the process is :
(1) Reversible (2) Spontaneous
(3) Exothermic (4) Endothermic
TD0062

58. Calculate the entropy of $Br_2(g)$ in the reaction $H_2(g) + Br_2(g) \rightarrow 2HBr(g)$, $\Delta S° = 20.1 JK^{-1}$ given, entropy of H_2 and HBr is 130.6 and 198.5 J mol^{-1} K^{-1} :-
(1) 246.3 JK^{-1} (2) 123.15 JK^{-1}
(3) 24.63 JK^{-1} (4) 20 KJK^{-1}
TD0063

59. In which of the following case entropy decreases–
(1) Solid changing to liquid
(2) Expansion of a gas
(3) Crystals dissolve
(4) Polymerisation
TD0065

60. Which of the following quantity is not zero for element in standard state :–
(1) Enthalpy of formation
(2) Entropy
(3) Free energy of formation
(4) All of these
TD0066

61. Entropy of an adiabatic reversible process is:-
(1) Positive (2) Zero
(3) Negative (4) Constant
TD0067

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GIBBS FREE ENERGY

62. A gas is allowed to expand under reversible adiabatic conditions then :-
(1) $\Delta G = 0$ (2) $\Delta T = 0$
(3) $\Delta S = 0$ (4) None of these
TD0068

63. For a reaction at 25^0C enthalpy change (ΔH) and entropy change (ΔS) are -11.7×10^3 $Jmol^{-1}$ and -105 J mol^{-1} K^{-1} respectively. The reaction is :
(1) Spontaneous (2) Non spontaneous
(3) At equilibrium (4) Can't say anything
TD0069

64. If $\Delta H > 0$ and $\Delta S > 0$, the reaction proceeds spontaneously when :-
(1) $\Delta H > 0$ (2) $\Delta H < T\Delta S$
(3) $\Delta H = T\Delta S$ (4) None
TD0071

65. The temperature at which the reaction
$Ag_2O(s) \longrightarrow 2Ag(s) + \frac{1}{2}O_2(g)$
is at equilibrium is ……;
Given $\Delta H = 30.5$ kJ mol^{-1}
and $\Delta S = 0.066$ kJK^{-1} mol^{-1} :
(1) 462.12 K (2) 362.12 K
(3) 262.12 K (4) 562.12 K
TD0072

66. Which of the following is true for the reaction $H_2O(\ell) \rightleftharpoons H_2O(g)$ at 100^0C and 1 atmosphere
(1) $\Delta S = 0$ (2) $\Delta H = 0$
(3) $\Delta H = \Delta E$ (4) $\Delta H = T\Delta S$
TD0074

67. For the reaction $A(s) \longrightarrow B(s) + C(g)$ the value of ΔH = 30.56 kJ mol^{-1} and ΔS=66 JK^{-1} mol^{-1}. The temperature at which the free energy change for the reaction will be zero is :–
(1) 373 K (2) 413 K
(3) 463 K (4) 493 K
TD0075

68. For hypothetical reversible reaction
$\frac{1}{2}A_2(g) + \frac{3}{2}B_2(g) \longrightarrow AB_3(g)$; $\Delta H = -20$ kJ if standard entropies of A_2, B_2 and AB_3 are 60, 40 and 50 JK^{-1} $mole^{-1}$ respectively. The above reaction will be in equilibrium at :–
(1) 400K (2) 500K
(3) 250K (4) 200K
TD0076

69. For the precipitation of AgCl by Ag^+ ions and HCl
(1) $\Delta H = 0$ (2) $\Delta G = 0$
(3) $\Delta G = -ve$ (4) $\Delta H = \Delta G$
TD0077

70. What is the sign of ΔG for the process of ice melting at 1 atm, 283 K is ?
(1) $\Delta G > 0$ (2) $\Delta G = 0$
(3) $\Delta G < 0$ (4) None of these
TD0078

71. A reaction $A + B \longrightarrow C + D + q$ is found to have a positive entropy change, the reaction will be -
(1) Possible at high temperature
(2) Possible only at low temperature
(3) Not possible at any temperature
(4) Possible at any temperature
TD0080

72. Equilibrium constant of a reaction is related to :
(1) Standard free energy change ΔG^0
(2) Free energy change ΔG
(3) Entropy change
(4) None
TD0081

73. The Vant Hoff equation is :
(1) $\Delta G° = RT \log_e K_P$ (2) $-\Delta G° = RT \log_e K_P$
(3) $\Delta G° = RT^2 \ell nK_P$ (4) None
TD0082

74. If $\Delta G^0 > 0$ for a reaction then :
(1) $K_P > 1$
(2) $K_P < 1$
(3) The products predominate in the equilibrium mixture
(4) None
TD0083

75. If the equilibrium constant for a reaction is 10, then the value of ΔG^0 will be
(R = 8JK^{-1} mol^{-1}, T = 300 K)
(1) + 5.527 kJ mol^{-1} (2) – 5.527 kJ mol^{-1}
(3) +55.27 kJ mol^{-1} (4) – 55.27 kJ mol^{-1}
TD0084

76. The process of evaporation of a liquid is accompanied by :
(1) Increase in enthalpy
(2) Decrease in free energy
(3) Increase in entropy
(4) All
TD0085

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77. For the process, $CO_2(s) \longrightarrow CO_2(g)$:
(1) Both ΔH and ΔS are +ve
(2) ΔH is negative and ΔS is +ve
(3) ΔH is +ve and ΔS is –ve
(4) Both ΔH and ΔS are –ve

TD0086

78. Which of the following provide exceptions to third law of thermodynamics
(1) CO (2) ice
(3) CO_2 (4) All the above

TD0087

79. The Gibbs free energy change of a reaction at 27°C is – 26 kCal. and its entropy change is – 60 Cals K. ΔH for the reaction is :–
(1) – 44 kCals. (2) – 18 kCals.
(3) 34 kals. (4) – 24 kCals.

TD0088

80. Which of the following reaction is expected never to be spontaneous :–
(1) $2O_3(g) \rightarrow 3O_2(g)$ $\Delta H = -Ve$, $\Delta S = +Ve$
(2) $Mg(s) + H_2(g) \rightarrow MgH_2$ $\Delta H = -Ve$, $\Delta S = -Ve$
(3) $Br_2(l) \rightarrow Br_2(g)$ $\Delta H = +Ve$, $\Delta S = +Ve$
(4) $2Ag(s) + 3N_2(g) \rightarrow 2AgN_3$ $\Delta H = +Ve$, $\Delta S = -Ve$

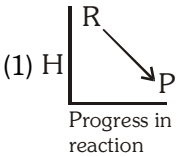
TD0089

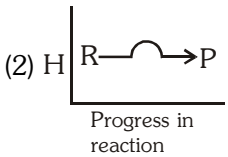
THERMOCHEMICAL REACTION

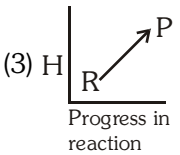
81. The formation of water from $H_2(g)$ and $O_2(g)$ is an exothermic process because :
(1) The chemical energy of $H_2(g)$ and $O_2(g)$ is more than that of water
(2) The chemical energy of $H_2(g)$ and $O_2(g)$ is less than that of water
(3) The temperature of $H_2(g)$ and $O_2(g)$ is higher than that of water
(4) The temperature of $H_2(g)$ and $O_2(g)$ is lower than that of water

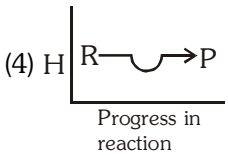
TC0090

82. Which plot represents for an exothermic reaction:

(1) 

(2) 

(3) 

(4) 

TC0091

83. Which one of the following is not applicable for a thermochemical equation :
(1) It tells about physical state of reactants and products
(2) It tells whether the reaction is spontaneous
(3) It tells whether the reaction is exothermic or endothermic
(4) It tells about the allotropic form (if any) of the reactants

TC0092

84. The correct thermochemical equation is :
(1) $C + O_2 \longrightarrow CO_2$; $\Delta H = -94$ kCal
(2) $C + O_2 \longrightarrow CO_2$; $\Delta H = +94.0$ kCal
(3) $C(s) + O_2(g) \longrightarrow CO_2(g)$; $\Delta H = -94$ kCal
(4) $C(s) + O_2(g) \longrightarrow CO_2(g)$; $\Delta H = +94$ kCal

TC0093

85. The enthalpy changes of formation of the gaseous oxide of nitrogen (N_2O and NO) are positive because of :
(1) The high bond energy of the nitrogen molecule
(2) The high electron affinity of oxygen atoms
(3) The high electron affinity of nitrogen atoms
(4) The tendency of oxygen to form O^{2-}

TC0094

86. ΔH for transition of carbon from diamond form to graphite form is – 453.5 Cal. This suggests that :
(1) Graphite is chemically different from diamond
(2) Graphite is as stable as diamond
(3) Graphite is more stable than diamond
(4) Diamond is more stable than graphite

TC0095

87. Which of the following values of heat of formation indicates that the product is least stable
(1) – 94 kCal (2) – 231.6 kCal
(3) + 21.4 kCal (4) + 64.8 kCal

TC0096

88. Heat of formation, ΔH_f° of an explosive compound like NCl_3 is –
(1) Positive (2) Negative
(3) Zero (4) Positive or negative

TC0097

89. According to the following reaction
$C(s) + 1/2\ O_2(g) \rightarrow CO(g)$, $\Delta H = -26.4$ kCal
(1) CO is an endothermic compound
(2) CO is an exothermic compound
(3) The reaction is endothermic
(4) None of the above

TC0098

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90. Which of the following represents an exothermic reaction:-
(1) $N_2(g) + O_2(g) \rightarrow 2NO(g)$, $\Delta H = 180.5$ kJ
(2) $H_2O(g) + C(s) \rightarrow CO(g) + H_2(g)$, $\Delta E = 131.2$kJ
(3) $2HgO(s) + 180.4$ KJ $\rightarrow 2Hg(\ell) + O_2(g)$
(4) $2Zn(s) + O_2(g) \rightarrow 2ZnO(s)$, $\Delta E = -693.8$ kJ

TC0099

91. Consider the reaction $3O_2 \rightarrow 2O_3$; $\Delta H = +Ve$, from the reaction, we can say that :–
(1) Ozone is more stable then oxygen
(2) Ozone is less stable then oxygen and ozone decomposes forming oxygen readily
(3) Oxygen is less stable than ozone and oxygen decomposes forming ozone readily

TC0101

(4) None of the above

92. From the reaction P(White) $\rightarrow$ P(Red) ; $\Delta H = -18.4$ kJ, it follows that :-
(1) Red P is readily formed from white P
(2) White P is readily formed from red P
(3) White P can not be converted to red p
(4) White P can be converted into red P and red P is more stable

TC0102

FACTORS AFFECTING HEAT OF REACTION

93. In Kirchoff's equation which factor affects the heat of reaction :
(1) Pressure (2) Temperature
(3) Volume (4) Atomicity

TC0103

94. The enthalpy of a reaction at 273 K is – 3.57 kJ. what will be the enthalpy of reaction at 373 K if ΔC_p = zero :–
(1) – 3.57 kJ (2) Zero
(3) $-3.57 \times \frac{373}{273}$ kJ (4) – 375 kJ

TC0105

95. For the reactions,
(i) $H_2(g) + Cl_2(g) \longrightarrow 2HCl(g) + xkJ$
(ii) $H_2(g) + Cl_2(g) \longrightarrow 2HCl(\ell) + ykJ$
Which one of the following statement is correct :
(1) x > y (2) x < y
(3) x = y (4) More data required

TC0106

HEAT OF FORMATION

96. Since the enthalpy of formation of the elements in their standard states is taken to be zero. The heat of formation (ΔH_f) of compounds :
(1) Is always negative
(2) Is always positive
(3) Is zero
(4) May be positive or negative

TC0107

97. Reaction $H_2(s) + I_2(g) \longrightarrow 2HI$; $\Delta H = 12.40$ kCal. According to this, heat of formation of HI will be –
(1) 12.40 kCal (2) – 12.40 kCal
(3) – 6.20 kCal (4) 6.20 kCal

TC0108

98. Enthalpy of a compound is equal to its :- (When it is formed from reference state of constituent elements)
(1) Heat of combustion
(2) Heat of formation
(3) Heat of reaction
(4) Heat of solution

TC0110

99. Which of the following equations respresents standard heat of formation of CH_4 ?
(1) $C_{(diamond)} + 2H_{2\,(g)} \longrightarrow CH_{4\,(g)}$
(2) $C_{(graphite)} + 2H_{2\,(g)} \longrightarrow CH_{4\,(g)}$
(3) $C_{(diamond)} + 4H_{(g)} \longrightarrow CH_{4\,(g)}$
(4) $C_{(graphite)} + 4H_{(g)} \longrightarrow CH_{4\,(g)}$

TC0111

100. The enthalpy of formation of ammonia is –46.0 kJ mol^{-1}. The enthalpy change for the reaction $2NH_3(g) \rightarrow N_2(g) + 3H_2(g)$ is :
(1) 46.0 kJ mol^{-1} (2) 92.0 kJ mol^{-1}
(3) – 23.0 kJ mol^{-1} (4) – 92.0 kJ mol^{-1}

TC0112

101. Given enthalpy of formation of $CO_2(g)$ and CaO(s) are – 94.0 kJ and – 152 kJ respectively and the enthalpy of the reaction :
$CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$ is 42 kJ. The enthalpy of formation of $CaCO_3(s)$ is
(1) – 42 KJ (2) – 202 KJ
(3) +202 KJ (4) – 288KJ

TC0113

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102. Given that standard heat enthalpy of CH_4, C_2H_4 and C_3H_8 are –17.9, 12.5, –24.8 kCal mol^{-1}. The ΔH for $CH_4 + C_2H_4 \rightarrow C_3H_8$ is :
(1) – 55.2 kCal (2) – 30.2 kCal
(3) 55.2 kCal (4) – 19.4 kCal
TC0114

103. The standard molar heat of formation of ethane, CO_2 and water(ℓ) are respectively –21.1, –94.1 and – 68.3 kCal. The standard molar heat of combustion of ethane will be
(1) –372 kCal (2) –162 kCal
(3) – 240 kCal (4) –183.5 kCal
TC0115

104. The ΔH_f° for $CO_{2(g)}$, $CO_{(g)}$ and $H_2O_{(g)}$ are –393.5, –110.5 and –241.8 kJ mol^{-1}respectively the standard enthalpy change (in kJ) for the reaction $CO_{2(g)}+H_{2(g)} \rightarrow CO_{(g)}+ H_2O_{(g)}$ is -
(1) 524.1 (2) 41.2 (3) –262.5 (4) –41.2
TC0117

105. The enthalpies of combustion of carbon and carbon monoxide are –393.5 kJ and –283 kJ, respectively the enthalpy of formation of carbon monoxide is :
(1) –676.5 kJ (2) –110.5 kJ
(3) 110.5 kJ (4) 676.5 kJ
TC0118

106. The standard heat of formation of $CS_2(\ell)$ will be; given that the standard heat of combustion of carbon (s), sulphur(s) and $CS_2(\ell)$ are –393.3, –293.72 and –1108.76 kJ mol^{-1} respectively is
(1) –128.02 kJ $mole^{-1}$
(2) +12.802 kJ mol^{-1}
(3) +128.02 kJ mol^{-1}
(4) –12.802 kJ mol^{-1}
TC0119

107. The heat of combustion of $CH_{4(g)}$, $C_{(s)}$ and $H_{2(g)}$ at 25 °C are –212.4 K Cal, –94.0 K Cal and –68.4 K Cal respectively, the heat of formation of CH_4 will be -
(1) +54.4 K Cal (2) –18.4 K Cal
(3) –375.2 K Cal (4) +212.8 K Cal
TC0120

108. Standard enthalpy of formation is zero for .
(1) $C_{diamond}$ (2) $Br_{(g)}$
(3) $C_{graphite}$ (4) $O_{3(g)}$
TC0121

109. The standard heats of formation of $NO_2(g)$ and $N_2O_4(g)$ are 8.0 and 2.0 kCal mol^{-1} respectively the heat of dimerization of NO_2 in kCal is
(1) 10.0 (2) –6.0 (3) –12.0 (4) –14.0
TC0122

110. M is a metal that forms an oxide M_2O

$$\frac{1}{2}M_2O \rightarrow M + \frac{1}{4}O_2 \quad \Delta H = 120 \text{ kCal.}$$

When a sample of metal M reacts with one mole of oxygen what will be the ΔH in that case
(1) 240 kCal. (2) – 240 kCal.
(3) 480 kCal. (4) – 480 kCal.
TC0123

HEAT OF COMBUSTION

111. According to equation,
$C_6H_6(\ell) + 15/2\ O_2(g) \longrightarrow 6CO_2(g) + 3H_2O(\ell)$; $\Delta H = -3264.4$ kJ mol^{-1} the energy evolved when 7.8 g benzene is burnt in air will be -
(1) 163.22 kJ (2) 32.64 kJ
(3) 3.264 kJ (4) 326.4 kJ
TC0124

112. Heat of combustion of CH_4,C_2H_6, C_2H_4 and C_2H_2 gases are –212.8, –373.0, –337.0 and –310.5 kCal respectively at the same temperature. The best fuel among these gases is :
(1) CH_4 (2) C_2H_6 (3) C_2H_4 (4) C_2H_2
TC0125

113. Given standard enthalpy of formation of CO (–110 kJ mol^{-1}) and CO_2(–394 kJ mol^{-1}). The heat of combustion when one mole of graphite burns is
(1) – 110 kJ (2) – 284 kJ
(3) – 394 kJ (4) – 504 kJ
TC0126

114. The enthalpy of formation for $C_2H_4(g)$, $CO_2(g)$ and $H_2O(\ell)$ at 25°C and 1 atm. pressure are 52, - 394 and - 286 kJ $mole^{-1}$ respectively. The enthalpy of combustion of C_2H_4 will be:-
(1) + 1412 kJ $mole^{-1}$
(2) –1412 kJ $mole^{-1}$
(3) + 142.2 kJ $mole^{-1}$
(4) –141.2 kJ $mole^{-1}$
TC0127

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115. The combustion of one mole of benzene takes place at 298 K and 1 atm. After combustion, $CO_2(g)$ and $H_2O(l)$ are produced and 3267.0 kJ of heat is liberated. Calculate the standard enthalpy of formation, $\Delta_f H^\ominus$ of benzene. Standard enthalpies of formation of $CO_2(g)$ and $H_2O(l)$ are –393.5 kJ mol^{-1} and –285.83 kJ mol^{-1} respectively.

(1) 48.51 kJ mol^{-1} (2) –48.51 kJ mol^{-1}

(3) –97.02 kJ mol^{-1} (4) 97.02 kJ mol^{-1}

TC0128

116. The heat evolved during the combustion of 112 litre of water gas at STP (mixture of equal volume of H_2 and CO) is : Given

$H_2(g) + ½ O_2(g) = H_2O(g)$; $\Delta H = -241.8$ kJ

$CO(g) + ½ O_2(g) = CO_2(g)$; $\Delta H = -283$ kJ

(1) 241.8 kJ (2) 283 kJ

(3) 1312 kJ (4) 1586 kJ

TC0130

117. A person requires 2870 kCal of energy to lead normal daily life. If heat of combustion of cane sugar is –1349 $kCalmol^{-1}$, then his daily consumption of sugar is :

(1) 728g (2) 0.728g

(3) 342g (4) 0.342g

TC0131

118. The following are the heats of reactions -

(i) ΔH_f° of $H_2O_{(l)} = -68.3$ kCal mol^{-1}

(ii) $\Delta H_{comb.}^\circ$ of $C_2H_2 = -337.2$ kCal mol^{-1}

(iii) $\Delta H_{comb.}^\circ$ of $C_2H_4 = -363.7$ kCal mol^{-1}

Then heat change for the reaction $C_2H_2 + H_2 \rightarrow C_2H_4$ is -

(1) –716.1 kCal (2) + 337.2 kCal

(3) –41.8 kCal (4) –579.5 kCal

TC0134

119. The heat of combustion of a substance is :-

(1) Always positive

(2) Always negative

(3) Numerically equal to the heat of formation

(4) 1 and 3 both

TC0135

120. The value of ΔH for the combustion of C(s) is –94.4 kCal. The heat of formation of $CO_2(g)$ is :-

(1) –49.5 kCal (2) –94.4 kCal

(3) –188.0 kCal (4) More data required

TC0136

121. In the combustion of 0.4 g. of CH_4, 0.25 kCal. of heat is liberated. The heat of combustion of CH_4 is

(1) – 20 kCal (2) – 10 kCal

(3) – 2.5 kCal (4) – 5 kCal

TC0137

122. If $C_6H_{12}O_6(s) + 9O_2(g) \rightarrow 6CO_2(g) + 6H_2O(g)$; ΔH= – 680 kCal The weight of $CO_2(g)$ produced when 170 kCal of heat is evolved in the combustion of glucose is:-

(1) 265 g (2) 66 g

(3) 11 g (4) 64 g

TC0138

123. Which of the following equations corresponds to the enthalpy of combustion at 298 K :-

(1) $C_2H_6(g) + 7/2\ O_2(g) \rightarrow 2CO_2(g) + 3H_2O(g)$

(2) $2C_2H_6(g) + 7\ O_2(g) \rightarrow 4CO_2(g) + 6H_2O(g)$

(3) $C_2H_6(g) + 7/2\ O_2(g) \rightarrow 2CO_2(g) + 3H_2O(\ell)$

(4) $2C_2H_6(g) + 7O_2(g) \rightarrow 4CO_2(g) + 6H_2O(\ell)$

TC0139

124. Heat of formation of CO_2 is - 94.0 kCal. What would be the quantity of heat liberated, when 3 g of graphite is burnt in excess of oxygen:-

(1) 23.5 kCal (2) 2.35 kCal

(3) 94.0 kCal (4) 31.3 kCal

TC0140

HEAT OF NEUTRALIZATION

125. The amount of heat liberated when one mole of NH_4OH reacts with one mole of HCl is

(1) 13.7 kCal (2) More than 13.7 kCal

(3) Less than 13.7 kCal (4) Cannot be predicted

TC0141

126. If $H^+ + OH^- = H_2O$ + 13.7 kCal, then heat of complete neutralisation of one gram mole of H_2SO_4 with strong base will be :

(1) –13.7 Kcal (2) –27.4 Kcal

(3) –6.85 Kcal (4) –3.425 KCal

TC0142

127. Heat of neutralisation of a strong dibasic acid in dilute solution by NaOH is nearly :

(1) – 27.4 kCal eq^{-1} (2) – 13.7 kCal eq^{-1}

(3) 13.7 kCal eq^{-1} (4) – 13.7 kCal mol^{-1}

TC0143

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128. The temperature of a 5 ml of strong acid increases by 5^0C when 5 ml of a strong base is added to it. If 10 ml of each are mixed temperature should increase by :

(1) 5^0 C (2) 10^0C

(3) 15^0C (4) Cannot be known

TC0144

129. The heat of neutralization of HCl by NaOH is – 55.9 kJ mol^{-1}. If the heat of neutralization of HCN by NaOH is – 12.1 kJ mol^{-1}. The energy of dissociation of HCN is

(1) – 43.8 kJ (2) 43.8 kJ

(3) 68 kJ (4) – 68 kJ

TC0145

130. If water is formed from H^+ ions and OH^- the heat exchange during the reaction :

(1) – 13.7 kCal (2) 13.7 kCal

(3) –63.4 kCal (4) More data required

TC0146

131. The change in the enthalpy of $NaOH + HCl \longrightarrow NaCl + H_2O$ is called :

(1) Heat of neutralisation

(2) Heat of reaction

(3) Heat of hydration

(4) Heat of solution

TC0147

HEAT OF HYDROGENATION

132. The heat of combustion of C_2H_4, C_2H_6 and H_2 are –1409.5 kJ, –1558.3 kJ and –285.6 kJ. The heat of hydrogenation of ethene is -

(1) –136.8 kJ (2) –13.68 kJ

(3) 273.6 kJ (4) 1.368 kJ

TC0149

133. The enthalpy of combustion of cyclohexane, cyclohexene and H_2 are respectively –3920, –3800 and –241 kJ mol^{-1}. The heat of hydrogenation of cyclohexene is:-

(1) –121 kJ mol^{-1} (2) 121 kJ mol^{-1}

(3) –242 kJ mol^{-1} (4) 242 kJ mol^{-1}

TC0150

BOND ENERGY/RESONANCE ENERGY

134. Bond energy of a molecule :

(1) Is always negative

(2) Is always positive

(3) Either positive or negative

(4) Depends upon the physical state of the system

TC0151

135. Among the following for which reaction heat of reaction represents bond energy of HCl

(1) $HCl(g) \longrightarrow H^+(g) + Cl^-(g)$

(2) $HCl(g) \longrightarrow \frac{1}{2}H_2(g) + \frac{1}{2}Cl_2(g)$

(3) $2HCl(g) \longrightarrow H_2(g) + Cl_2(g)$

(4) $HCl(g) \longrightarrow H(g) + Cl(g)$

TC0152

136. The bond energies of F_2, Cl_2, Br_2 and I_2 are 155.4, 243.6, 193.2 and 151.2 kJ mol^{-1} respectively. The strongest bond is :

(1) F – F (2) Cl – Cl

(3) Br – Br (4) I – I

TC0153

137. Energy required to dissociate 4g of gaseous hydrogen into free gaseous atoms is 208 kCal at 25^0C. The bond energy of H—H bond will be :

(1) 1.04 kCal (2) 10.4 kCal

(3) 104 kCal (4) 1040 kCal

TC0154

138. Heat evolved in the reaction $H_2 + Cl_2 \longrightarrow 2HCl$ is 182 kJ. Bond energies of H–H and Cl–Cl are 430 and 242 kJ mol^{-1} respectively. The H–Cl bond energy is :

(1) 245 kJ mol^{-1} (2) 427 kJ mol^{-1}

(3) 336 kJ mol^{-1} (4) 154 kJ mol^{-1}

TC0155

139. The enthalpy change for the reaction $H_2(g) + C_2H_4(g) \rightarrow C_2H_6(g)$ is..... . The bond energies are in kCal mol^{-1}

H – H = 103, C – H = 99, C – C = 80 & C = C =145

(1) –10 kCal mol^{-1}

(2) +10 kCal mol^{-1}

(3) – 30 kCal mol^{-1}

(4) +30 kCal mol^{-1}

TC0156

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140. Bond dissociation enthalphies of $H_2(g)$ and $N_2(g)$ are 436.0 kJ mol^{-1} and 941.8 kJ mol^{-1} respectively and enthalpy of formation of $NH_3(g)$ is –46 kJ mol^{-1}. What is enthalpy of atomization of $NH_3(g)$?

(1) 390.3 kJ mol^{-1} (2) 1170.9 kJ mol^{-1}

(3) 590 kJ mol^{-1} (4) 720 kJ mol^{-1}

TC0157

141. From the reactions :

$C(s) + 2H_2(g) \rightarrow CH_4(g)\ \Delta H = - X$ Kcal

$C(g) + 4H(g) \rightarrow CH_4(g),\ \Delta H = - X_1$ Kcal

$CH_4(g) \rightarrow CH_3(g) + H(g)\ \Delta H = + Y$(Kcal)

Bond energy of C–H bond is –

(1) $\frac{X}{4}$ kCal mol^{-1} (2) Y kCal mol^{-1}

(3) $\frac{X_1}{4}$ kCal mol^{-1} (4) X_1 kCal mol^{-1}

TC0158

142. The enthalpy changes at 298 K in successive breaking of O–H bonds of water are

$H_2O \longrightarrow H(g) + OH(g);\ \Delta H = 498$ kJ mol^{-1}

$OH(g) \longrightarrow H(g) + O(g)\ ;\ \Delta H = 428$ kJmol^{-1}

the bond enthalpy of O–H bond is

(1) 498 kJ mol^{-1} (2) 428 kJ mol^{-1}

(3) 70 kJ mol^{-1} (4) 463 kJ mol^{-1}

TC0159

143. If ΔH_f° of $ICl_{(g)}$, $Cl_{(g)}$, and $I_{(g)}$ is 17.57, 121.34 and 106.96 J mol^{-1} respectively. Then bond dissociation energy of ICl bond is -

(1) 35.15 J mol^{-1} (2) 106.69 J mol^{-1}

(3) 210.73 J mol^{-1} (4) 420.9 J mol^{-1}

TC0160

144. Heat of dissociation of benzene to elements is 5535 kJ mol^{-1}. The bond enthalpies of C–C, C=C and C–H are 347.3, 615.0 and 416.2 kJ respectively. Magnitude of resonance energy of benzene is

(1) 1.51 kJ (2) 15.1 kJ

(3) 151 kJ (4) 1511 kJ

TC0161

SOME OTHER HEAT OF REACTIONS

145. The enthalpy change for the reaction

$2C(graphite) + 3H_2(g) \longrightarrow C_2H_6(g)$ is called

(1) Enthalpy of formation

(2) Enthalpy of combustion

(3) Enthalpy of hydrogenation

(4) Enthalpy of vaporisation

TC0162

146. $Cl_2(g) \longrightarrow 2Cl(g)$, In this process value of ΔH will be -

(1) Positive

(2) Negative

(3) Zero

(4) Nothing can be predicted

TC0163

147. The magnitude of heat of solution ..... on addition of solvent to solution

(1) Decreases

(2) Increases

(3) Remains constant

(4) Increases or decreases

TC0164

148. If $H_2(g) \longrightarrow 2H(g)\ ;\ \Delta H = 104$ kCal, than heat of atomisation of hydrogen is :

(1) 52 kCal (2) 104 kCal

(3) 208 kCal (4) None of these

TC0165

149. $S_{(rhombic)} + O_{2(g)} \longrightarrow SO_{2(g)}\ ;\ \Delta H = -297.5$ kJ

$S_{(monoclinic)} + O_{2\ (g)} \longrightarrow SO_2\ ;\ \Delta H = -300$ kJ

The data can predict that –

(1) Rhombic sulphur is yellow in colour

(2) Monoclinic sulphur has metallic lusture.

(3) Monoclinic sulphur is more stable

(4) ΔH transition of S_R to S_M is endothermic

TC0166

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150. The heat of combustion of yellow phosphorous and red phosphorous are – 9.91 kJ and –8.78 kJ respectively. The heat of transition of yellow phosphorous to red phosphorous is

(1) –18.69 kJ (2) +1.13 kJ

(3) +18.69 kJ (4) –1.13 kJ

TC0167

151. $2CO_{(g)} + O_{2(g)} \longrightarrow 2CO_{2(g)} + X$ kJ

In the above equation X kJ refers to :

(1) Heat of formation of CO_2

(2) Heat of vapourisation

(3) Heat of reaction

(4) Heat of sublimation

TC0169

152. ΔH for the reaction, $I_{(g)} + I_{(g)} \rightarrow I_2(g)$ will be:-

(1) Zero (2) – ve (3) + ve (4) ∞

TC0171

153. Given that :

$A(s) \rightarrow A(\ell)$; $\Delta H = x$, $A(\ell) \rightarrow A(g)$ - y

The heat of sublimation of A will be:-

(1) x + y (2) x - y

(3) x or y (4) - (x + y)

TC0172

HESS LAW

154. The enthalpy change of a reaction does not depend on

(1) State of reactants and products

(2) Nature of reactants and products

(3) Different intermediate reactions

(4) Initial and final enthalpy change of reaction

TC0173

155. From the thermochemical reactions,

$C(\text{graphite}) + \frac{1}{2}O_2 \longrightarrow CO$; $\Delta H = -110.5$ kJ

$CO + \frac{1}{2}O_2 \longrightarrow CO_2$; $\Delta H = -283.2$ kJ

the heat of reaction of $C(\text{graphite}) + O_2 \longrightarrow CO_2$ is :

(1) 393.7 kJ (2) – 393.7 kJ

(3) – 172.7 kJ (4) + 172.7 kJ

TC0174

156. If $H_2 + \frac{1}{2}O_2 \longrightarrow H_2O$; $\Delta H = -68.39$ kCal

$K + H_2O + \text{water} \longrightarrow KOH(aq) + \frac{1}{2}H_2$; $\Delta H = -48.0$ kCal

$KOH + \text{water} \longrightarrow KOH\ (aq)$ $\Delta H = -14.0$ kCal

the heat of formation of KOH is -

(1) – 68.39 + 48 – 14.0

(2) – 68.39 – 48.0 +14.0

(3) +68.39 – 48.0 + 14.0

(4) + 68.39 + 48.0 – 14.0

TC0175

157. Given $C(s) + O_2(g) \longrightarrow CO_2(g) + 94.2$ kCal

$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(\ell) + 68.3$ kCal

$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(\ell) + 210.8$ kCal

The heat of formation of methane in Kcal will be

(1) –45.9 (2) –47.8

(3) –20.0 (4) –47.3

TC0176

158. If, $H_2(g) + Cl_2(g) \longrightarrow 2HCl(g)$; $\Delta H^0 = -44$ kCal

$2Na(s) + 2HCl(g) \longrightarrow 2NaCl(s) + H_2(g)$;

$\Delta H = -152$ kCal

Then, $Na(s) + 0.5\ Cl_2(g) \longrightarrow NaCl(s)$; $\Delta H^0 = ?$

(1) 108 kCal (2) 196 kCal

(3) – 98 kCal (4) 54 kCal

TC0178

159. (i) $S(s) + 3/2\ O_2(g) \rightarrow SO_3(g) + 2x$ kCal

(ii) $SO_2(g) + \frac{1}{2}O_2(g) \rightarrow SO_3(g) + y$ kCal

Calculate the heat of formation of SO_2 :

(1) (2x + y) (2) –(2x – y)

(3) x + y (4) 2x / y

TC0179

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160. If $S + O_2 \longrightarrow SO_2$; $\Delta H = -298.2$

$SO_2 + \frac{1}{2} O_2 \longrightarrow SO_3$; $\Delta H = -98.7$

$SO_3 + H_2O \longrightarrow H_2SO_4$; $\Delta H = -130.2$

$H_2 + \frac{1}{2} O_2 \longrightarrow H_2O$; $\Delta H = -287.3$

Then the enthalpy of formation of H_2SO_4 at 298 K is -

(1) –814.4 kJ (2) –650.3 kJ

(3) –320.5 kJ (4) –433.5 kJ

TC0180

161. Given that :

$Zn + \frac{1}{2} O_2 \rightarrow ZnO + 84000$ Cal 1

$Hg + \frac{1}{2} O_2 \rightarrow HgO + 21700$ Cal 2

The heat of reaction (ΔH) for,

$Zn + HgO \rightarrow ZnO + Hg$ is :-

(1) 105700 Cal (2) 62300 Cal

(3) –105700 Cal (4) – 62300 Cal

TC0181

162. Given that -

$2C(s) + 2O_2(g) \rightarrow 2CO_2(g)$ $\Delta H = -787$ kJ

$H_2(g) + \frac{1}{2} O_2(g) \rightarrow H_2O(\ell)$ $\Delta H = -286$ kJ

$C_2H_2(g) + \frac{5}{2} O_2(g) \rightarrow 2CO_2(g) + H_2O(\ell) \Delta H = -1310$ kJ

Heat of formation of acetylene is :-

(1) + 1802 kJ (2) – 1802 kJ

(3) – 800 kJ (4) + 237 kJ

TC0182

163. The heat of reaction for

$A + \frac{1}{2} O_2 \rightarrow AO$ is - 50 kCal and

$AO + \frac{1}{2} O_2 \rightarrow AO_2$ is 100 kCal. The heat of reaction for $A + O_2 \rightarrow AO_2$ is:-

(1) – 50 kCal (2) + 50 kCal

(3) 100 kCal (4) 150 kCal

TC0184

164. $C(s) + O_2(g) \rightarrow CO_2(g) + 94.0$ kCal

$CO(g) + \frac{1}{2} O_2(g) \rightarrow CO_2(g)$, $\Delta H = -67.7$ kCal

from the above reactions find how much heat (kCal $mole^{-1}$) would be produced in the following reaction: $C(s) + \frac{1}{2} O_2(g) \rightarrow CO(g)$

(1) 20.6 (2) 26.3

(3) 44.2 (4) 161.6

TC0185

165. The enthalpy of vapourisation of liquid water using the data:

$H_2(g) + \frac{1}{2} O_2(g) \rightarrow H_2O(\ell)$; $\Delta H = -285.77$ kJ mol^{-1}

$H_2(g) + \frac{1}{2} O_2(g) \rightarrow H_2O(g)$; $\Delta H = -241.84$ kJ mol^{-1}

(1) + 43.93 kJ mol^{-1} (2) - 43.93 kJ mol^{-1}

(3) + 527.61 kJ mol^{-1} (4) - 527.61 kJ mol^{-1}

TC0187

166. $H_2(g) + \frac{1}{2} O_2(g) = H_2O(\ell)$; $\Delta H_{298K} = -68.32$kCal.

Heat of vapourisation of water at 1 atm and 25^0C is 10.52 kCal. The standard heat of formation (in kCal) of 1 mole of water vapour at 25^0C is

(1) 10.52 (2) –78.84

(3) +57.80 (4) –57.80

TC0188

167. The heat of solution of anhydrous $CuSO_4$ and $CuSO_4.5H_2O$ are – 15.89 and 2.80 kCal mol^{-1} respectively. What will be the heat of hydration of anhydrous $CuSO_4$?

(1) –18.69 kCal (2) 18.69 kCal

(3) –28.96 kCal (4) 28.96 kCal

TC0189

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168. Which of the following expressions is true:-

(1) H_f^0 (CO,g) = $\frac{1}{2}\Delta H_f^0$ (CO_2,g)

(2) ΔH_f^0 (CO,g)=ΔH_f^0 (C,graphite) + $\frac{1}{2}$ ΔH_f^0(O_2,g)

(3) ΔH_f^0 (CO,g)=ΔH_f^0 (CO_2,g) - $\frac{1}{2}$ ΔH_f^0 (O_2,g)

(4) ΔH_f^0 (CO,g)=ΔH_{comb}^0 (C,graphite)-ΔH_{comb}^0 (CO,g)

TC0191

EXERCISE-I (Conceptual Questions) ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**2**	**3**	**4**	**4**	**3**	**3**	**1**	**4**	**1**	**1**	**1**	**4**	**2**	**1**	**4**
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	**4**	**4**	**1**	**1**	**3**	**1**	**3**	**2**	**2**	**4**	**2**	**2**	**4**	**2**	**1**
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	**1**	**1**	**2**	**3**	**4**	**2**	**2**	**2**	**3**	**1**	**1**	**1**	**2**	**3**	**4**
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	**3**	**4**	**3**	**1**	**1**	**2**	**3**	**3**	**1**	**3**	**2**	**2**	**1**	**4**	**2**
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	**4**	**3**	**2**	**2**	**1**	**4**	**3**	**2**	**3**	**3**	**4**	**1**	**2**	**2**	**2**
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	**4**	**1**	**4**	**1**	**4**	**1**	**1**	**2**	**3**	**1**	**3**	**4**	**1**	**2**	**4**
Que.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Ans.	**2**	**4**	**2**	**1**	**2**	**4**	**4**	**2**	**2**	**2**	**4**	**4**	**1**	**2**	**2**
Que.	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Ans.	**3**	**2**	**3**	**4**	**4**	**4**	**1**	**3**	**2**	**2**	**3**	**1**	**3**	**2**	**2**
Que.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
Ans.	**2**	**2**	**3**	**1**	**3**	**2**	**2**	**1**	**2**	**1**	**1**	**1**	**1**	**2**	**4**
Que.	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
Ans.	**2**	**3**	**2**	**3**	**2**	**3**	**4**	**3**	**3**	**1**	**1**	**3**	**2**	**4**	**4**
Que.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165
Ans.	**3**	**2**	**1**	**3**	**2**	**2**	**3**	**3**	**2**	**1**	**4**	**4**	**2**	**2**	**1**
Que.	166	167	168												
Ans.	**4**	**1**	**4**												

EXERCISE-III (Analytical Questions)

Master Your Understanding

1. Consider the reaction : $N_2 + 3H_2 \rightarrow 2NH_3$ carried out at constant temperature and pressure. If ΔH and ΔU are the enthalpy and internal energy changes for the reaction, which of the following expressions is true ?

(1) $\Delta H < \Delta U$ (2) $\Delta H > \Delta U$

(3) $\Delta H = 0$ (4) $\Delta H = \Delta U$

TH0258

2. For a reversible process at T = 300K, the volume is increased from V_i = 1L to V_f = 10L. Calculate ΔH if the process is isothermal -

(1) 11.47 kJ (2) 4.98 kJ

(3) 0 (4) –11.47 kJ

TH0259

3. A piston filled with 0.04 mol of an ideal gas expands reversibly from 50.0 mL to 375 mL at a constant temperature of 37.0°C. As it does so, it absorbs 208 J of heat. The values of q and w for the process will be :-

(R = 8.314 J mol^{-1} K^{-1}) (ln 7.5 = 2.01)

(1) q = + 208 J, w = – 208 J

(2) q = – 208 J, w = – 208 J

(3) q = – 208 J, w = + 208 J

(4) q = + 208 J, w = + 208 J

TH0261

4. The entropy change involved in the isothermal reversible expansion of 2 moles of an ideal gas from a volume of 10 dm^3 to a volume of 100 dm^3 at 27°C is :-

(1) 32.3 J mol^{-1} K^{-1} (2) 42.3 J mol^{-1} K^{-1}

(3) 38.3 J mol^{-1} K^{-1} (4) 35.8 J mol^{-1} K^{-1}

TD0262

5. The value of enthalpy change (ΔH) for the reaction $C_2H_5OH_{(\ell)} + 3O_{2(g)} \rightarrow 2CO_{2(g)} + 3H_2O_{(\ell)}$ at 27°C is –1366.5 kJ mol^{-1}. The value of internal energy change for the above reaction at this temperature will be :-

(1) –1371.5 kJ (2) –1369.0 kJ

(3) –1364.0 kJ (4) –1361.5 kJ

TH0263

6. For the process $H_2O(\ell)$ (1 bar , 373K) $\rightarrow$ $H_2O(g)$ (1 bar , 373K), the correct set of thermodynamic parameters is :

(1) $\Delta G = 0$, $\Delta S = +ve$

(2) $\Delta G = 0$, $\Delta S = -ve$

(3) $\Delta G = +ve$, $\Delta S = 0$

(4) $\Delta G = -ve$, $\Delta S = +ve$

TD0264

7. Which of the following pairs of a chemical reaction is certain to result in a spontaneous reaction.?

(1) endothermic and decreasing disorder

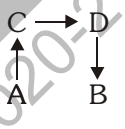
(2) exothermic and increasing disorder

(3) endothermic and increasing disorder

(4) exothermic and decreasing disorder

TD0265

8. The conversion A to B is carried out by the following path :



Given : $\Delta S_{(A \rightarrow C)}$ = 50 e.u. , $\Delta S_{(C \rightarrow D)}$ = 30 e.u., $\Delta S_{(B \rightarrow D)}$ = 20 e.u.

where e.u. is entropy unit then $\Delta S_{(A \rightarrow B)}$ is

(1) + 100 e.u.

(2) + 60 e.u.

(3) – 100 e.u.

(4) – 60 e.u.

TD0266

9. In a fuel cell methanol is used as fuel and oxygen gas is used as an oxidizer. The reaction is

$$CH_3OH(\ell) + \frac{3}{2}O_2(g) \longrightarrow CO_2(g) + 2H_2O(\ell)$$

At 298 K standard Gibb's energies of formation for $CH_3OH(\ell)$, $H_2O(\ell)$ and $CO_2(g)$ are –166.2, –237.2 and –394.4 kJ mol^{-1} respectively. If standard enthalpy of combustion of methanol is –726 kJ mol^{-1}, efficiency of the fuel cell will be

(1) 90% (2) 97% (3) 80% (4) 87%

TC0268

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10. Identify the correct statement regarding a sponateous process :-
(1) For a spontaneous process in an isolated system, the change in total entropy is positive
(2) Endothermic processes are never spontaneous
(3) Exothermic processes are always spontaneous
(4) Lowering of energy in the reaction process is the only criterion for spontaneity

TD0269

11. Standard entropy of X_2, Y_2 and XY_3 are 60, 40 and 50 JK^{-1} mol^{-1}, respectively. For the reaction,

$\frac{1}{2}X_2 + \frac{3}{2}Y_2 \rightarrow XY_3$, $\Delta H = -50$ kJ, to be at equilibrium, the temperature will be
(1) 1250 K (2) 500 K
(3) 750 K (4) 1000 K

TD0270

12. Consider the reaction :
$4NO_{2(g)} + O_{2(g)} \rightarrow 2N_2O_{5(g)}$, $\Delta_rH = -111$kJ.
If $N_2O_{5(s)}$ is formed instead of $N_2O_{5(g)}$ in the above reaction, the Δ_rH value will be :-
(given, ΔH of sublimation for N_2O_5 is 54 kJ mol^{-1})
(1) –165 kJ (2) +54 kJ
(3) +219 kJ (4) –219 kJ

TC0273

13. For complete combustion of ethanol, $C_2H_5OH(\ell) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(\ell)$, the amount of heat produced as measured in bomb calorimeter, is 1364.47 kJ mol^{-1} at 25°C. Assuming ideality the Enthalpy of combustion, Δ_cH, for the reaction will be :- (R = 8.314 kJ mol^{-1})
(1) – 1460.50 kJ mol^{-1}
(2) – 1350.50 kJ mol^{-1}
(3) – 1366.95 kJ mol^{-1}
(4) – 1361.95 kJ mol^{-1}

TH0274

14. If the bond dissociation energies of XY, X_2 and Y_2 (all diatomic molecules) are in the ratio of 1 : 1 : 0.5 and Δ_fH for the formation of XY is –400 kJ mol^{-1}. The bond dissociation energy of X_2 will be :-
(1) 200 kJ mol^{-1} (2) 100 kJ mol^{-1}
(3) 1600 kJ mol^{-1} (4) 300 kJ mol^{-1}

TC0275

15. The standard enthlapy of formation ($\Delta_fH°$) at 298K for methane, $CH_4(g)$, is –74.8 kJ mol^{-1}. The additional information required to determine the average energy for C–H bond formation would be:-
(1) Latent heat of vapourization of methane
(2) The first four ionization energies of carbon and electron gain enthalpy of hydrogen
(3) The dissociation energy of hydrogen molecule H_2
(4) The dissociation energy of H_2 and enthalpy of sublimation of carbon

TC0276

16. The incorrect expression among the following is :-
(1) $K = e^{-\Delta G°/RT}$
(2) $\frac{\Delta G_{system}}{\Delta S_{total}} = -T$
(3) In isothermal process, $W_{reversible} = -nRT \ln \frac{V_f}{V_i}$
(4) $\ln K = \frac{\Delta H° - T\Delta S°}{RT}$

TD0278

17. On the basis of the following thermochemical data : $\left[\Delta H_f^0\left(H^+_{(aq)} = 0\right)\right]$

$H_2O(\ell) \rightarrow H^+(aq) + OH^-(aq)$; $\Delta H = 57.32$ kJ

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(\ell)$; $\Delta H = -286.20$ kJ

The value of enthalpy of formation of OH^- ion at 25°C is :-
(1) +228.88 kJ (2) –343.52 kJ
(3) –22.88 kJ (4) –228.88 kJ

TC0280

18. Identify the correct statement for change of Gibbs energy for a system (ΔG_{system}) at constant temperature and pressure.
(1) If $\Delta G_{system} > 0$, the process is spontaneous.
(2) If $\Delta G_{system} = 0$, the system has attained equilibrium.
(3) If $\Delta G_{system} = 0$, the system is still moving in a particular direction.
(4) If $\Delta G_{system} < 0$, the process is not spontaneous.

TH0192

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19. Assume each reaction is carried out in an open container. For which reaction will $\Delta H = \Delta E$?
(1) $H_2(g) + Br_2(g) \rightarrow 2\,HBr(g)$
(2) $C(s) + 2\,H_2O(g) \rightarrow 2\,H_2(g) + CO_2(g)$
(3) $PCl_5(g) \rightarrow PCl_3(g) + Cl_2(g)$
(4) $2\,CO(g) + O_2(g) \rightarrow 2\,CO_2(g)$

TH0193

20. The enthalpy and entropy change for reaction $Br_2(\ell) + Cl_2(g) \rightarrow 2\,BrCl(g)$ are 30 kJ mol^{-1} and 105 JK^{-1} mol^{-1} respectively. The temperature at which the reaction will be in equilibrium is
(1) 285.7 K (2) 273 K
(3) 450 K (4) 300 K

TD0194

21. The enthalpy of hydrogenation of cyclohexene is –119.5 kJ mol^{-1}. If resonance energy of benzene is –150.4 kJ mol^{-1}, its enthalpy of hydrogenation would be
(1) – 508.9 kJ mol^{-1} (2) – 208.1 kJ mol^{-1}
(3) – 269.9 kJ mol^{-1} (4) – 358.5 kJ mol^{-1}

TC0195

22. Consider the following reactions :
(a) $H^+_{(aq)} + OH^-_{(aq)} = H_2O_{(l)}$, $\Delta H = -X_1$ kJ mol^{-1}

(b) $H_{2(g)} + \frac{1}{2}O_{2(g)} = H_2O_{(l)}$, $\Delta H = -X_2$ kJ mol^{-1}

(c) $CO_{2(g)} + H_{2(g)} = CO_{(g)} + H_2O_{(l)} - X_3$ kJ mol^{-1}

(d) $C_2H_{2(g)} + \frac{5}{2}O_{2(g)} = 2CO_{2(g)} + H_2O_{(l)} + X_4$ kJ mol^{-1}

Enthalpy of formation of $H_2O_{(l)}$ is :
(1) $+X_1$ kJ mol^{-1} (2) $-X_2$ kJ mol^{-1}
(3) $+X_3$ kJ mol^{-1} (4) $-X_4$ kJ mol^{-1}

TC0200

23. Given that bond energies of H–H and Cl–Cl are 430 kJ mol^{-1} and 240 kJ mol^{-1} respectively and $\Delta_f H$ for HCl is –90 kJ mol^{-1}. Bond enthalpy of HCl is :
(1) 245 kJ mol^{-1} (2) 290 kJ mol^{-1}
(3) 380 kJ mol^{-1} (4) 425 kJ mol^{-1}

TC0201

24. Bond dissociation enthalpy of H_2, Cl_2 and HCl are 434, 242 and 431 kJmol^{-1} respectively. Enthalpy of formation of HCl is:-
(1) –93 kJmol^{-1} (2) 245 kJmol^{-1}
(3) 93 kJmol^{-1} (4) –245 kJmol^{-1}

TC0202

25. For the gas phase reaction,

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

Which of the following conditions are correct ?
(1) $\Delta H < 0$ and $\Delta S < 0$ (2) $\Delta H > 0$ and $\Delta S < 0$
(3) $\Delta H = 0$ and $\Delta S < 0$ (4) $\Delta H > 0$ and $\Delta S > 0$

TD0203

26. Which of the following does not depends on path?
(I) q + w (II) q
(III) w (IV) H-TS
(1) (I),(II) and (III) (2) (II) and (III)
(3) (I) and (IV) (4) (II), (III) and (IV)

TH0204

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EXERCISE-III (Analytical Questions) ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**1**	**3**	**1**	**3**	**3**	**1**	**2**	**2**	**2**	**1**	**1**	**4**	**3**	**3**	**4**
Que.	16	17	18	19	20	21	22	23	24	25	26				
Ans.	**4**	**4**	**2**	**1**	**1**	**2**	**2**	**4**	**1**	**4**	**3**				